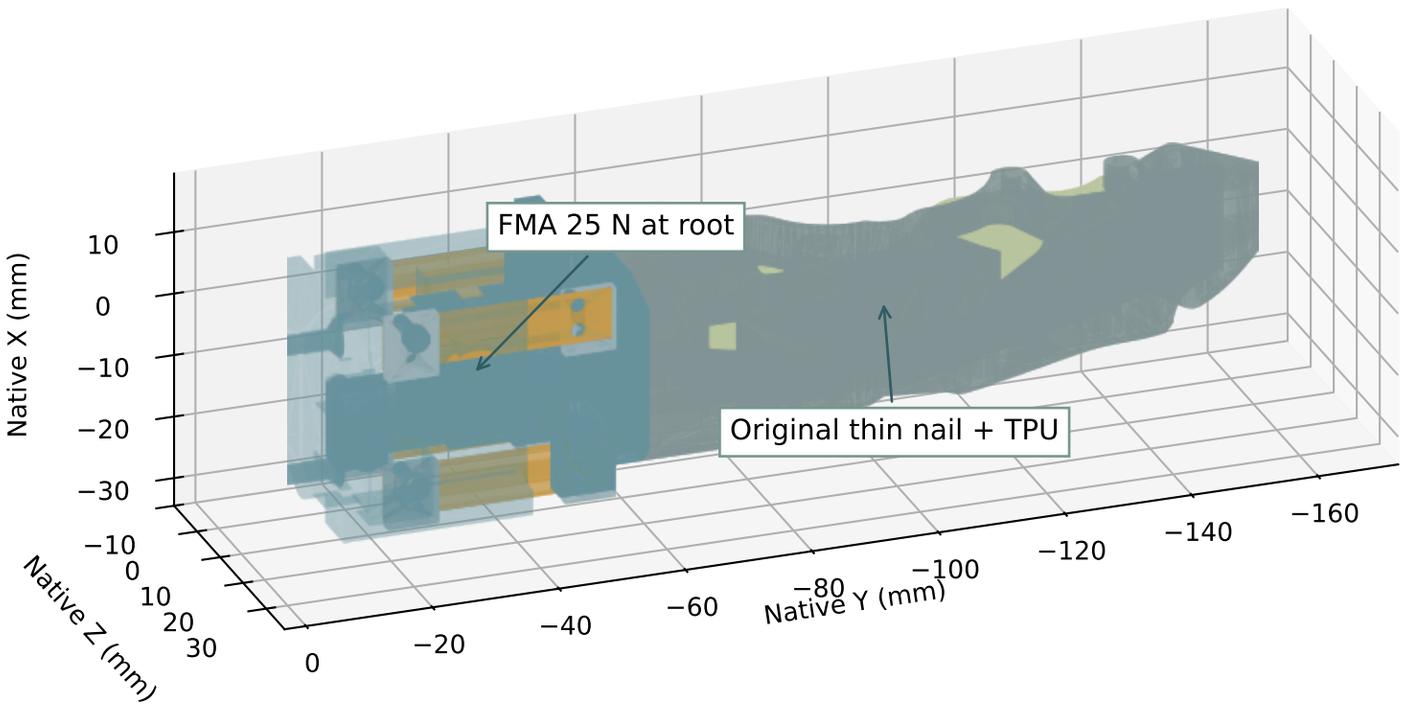
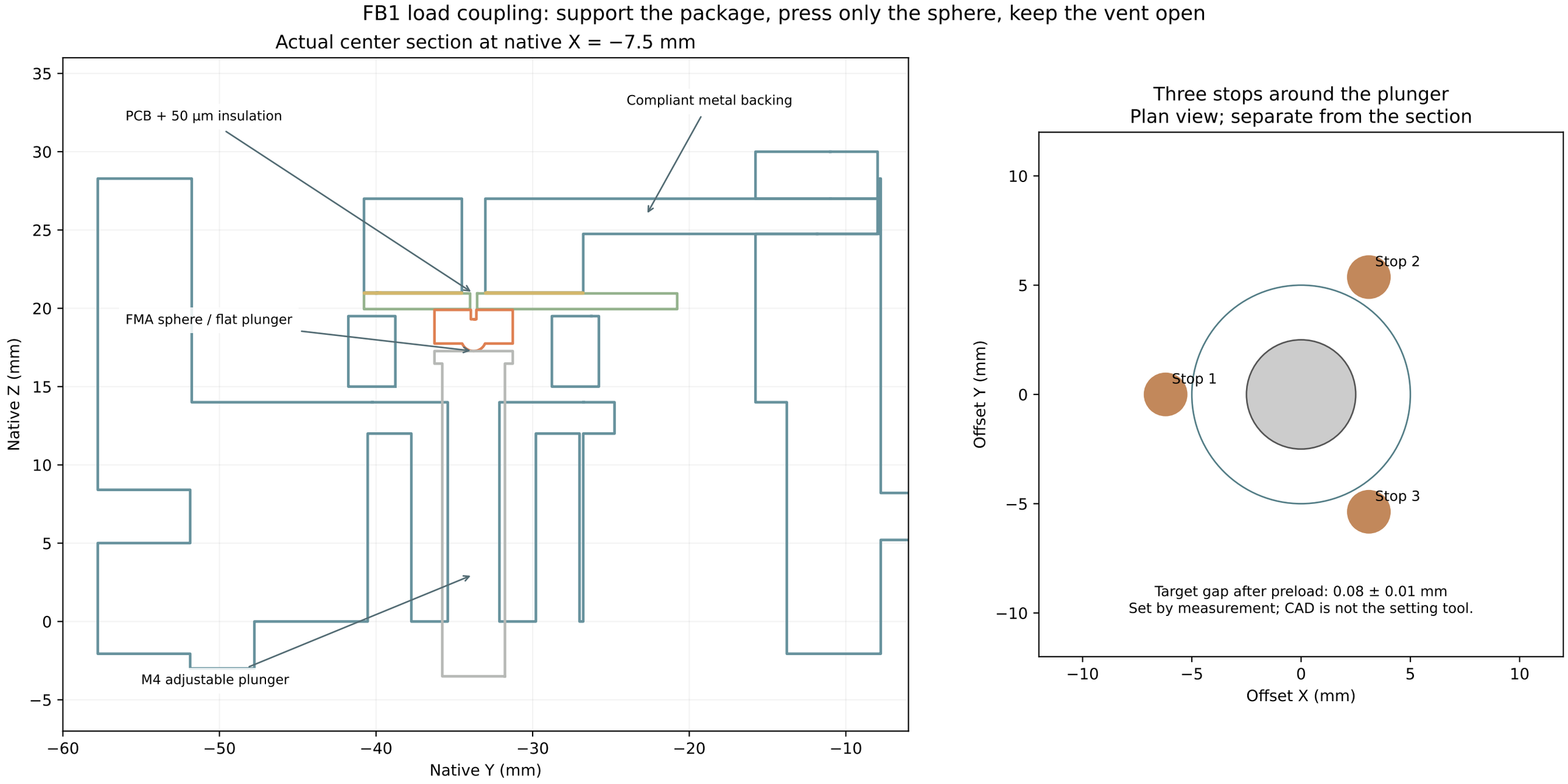
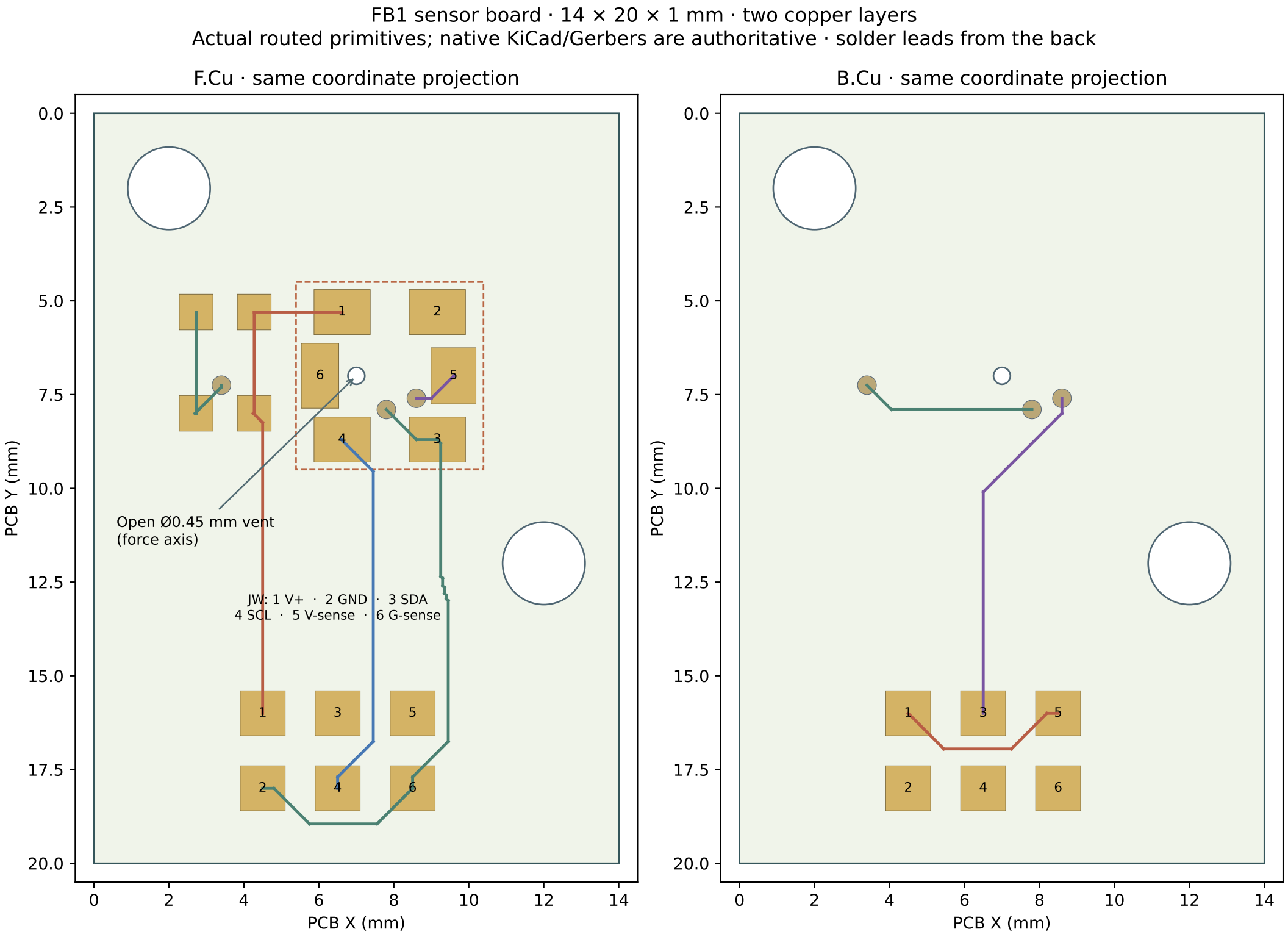
FB1: the complete finger loads a packaged sensor at the root
50 mm added reach · no sensor film on the leading tip · CAD rendering, not a physical specimen




FB1 load coupling: support the package, press only the sphere, keep the vent open
Actual center section at native X = −7.5 mm
PCB + 50 µm insulation
Compliant metal backing
FMA sphere / flat plunger
M4 adjustable plunger
Native Z (mm)
Native Y (mm)
Three stops around the plunger
Plan view; separate from the section
Stop 1
Stop 2
Stop 3
Target gap after preload: 0.08 ± 0.01 mm
Set by measurement; CAD is not the setting tool.
Offset X (mm)
Offset Y (mm)


FB1 sensor board · 14 × 20 × 1 mm · two copper layers
Actual routed primitives; native KiCad/Gerbers are authoritative · solder leads from the back
F.Cu · same coordinate projection
B.Cu · same coordinate projection
PCB Y (mm)
PCB X (mm)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
2
4
6
8
10
12
14
Open Ø0.45 mm vent
(force axis)
JW: 1 V+ · 2 GND · 3 SDA
4 SCL · 5 V-sense · 6 G-sense
1
2
3
4
5
6